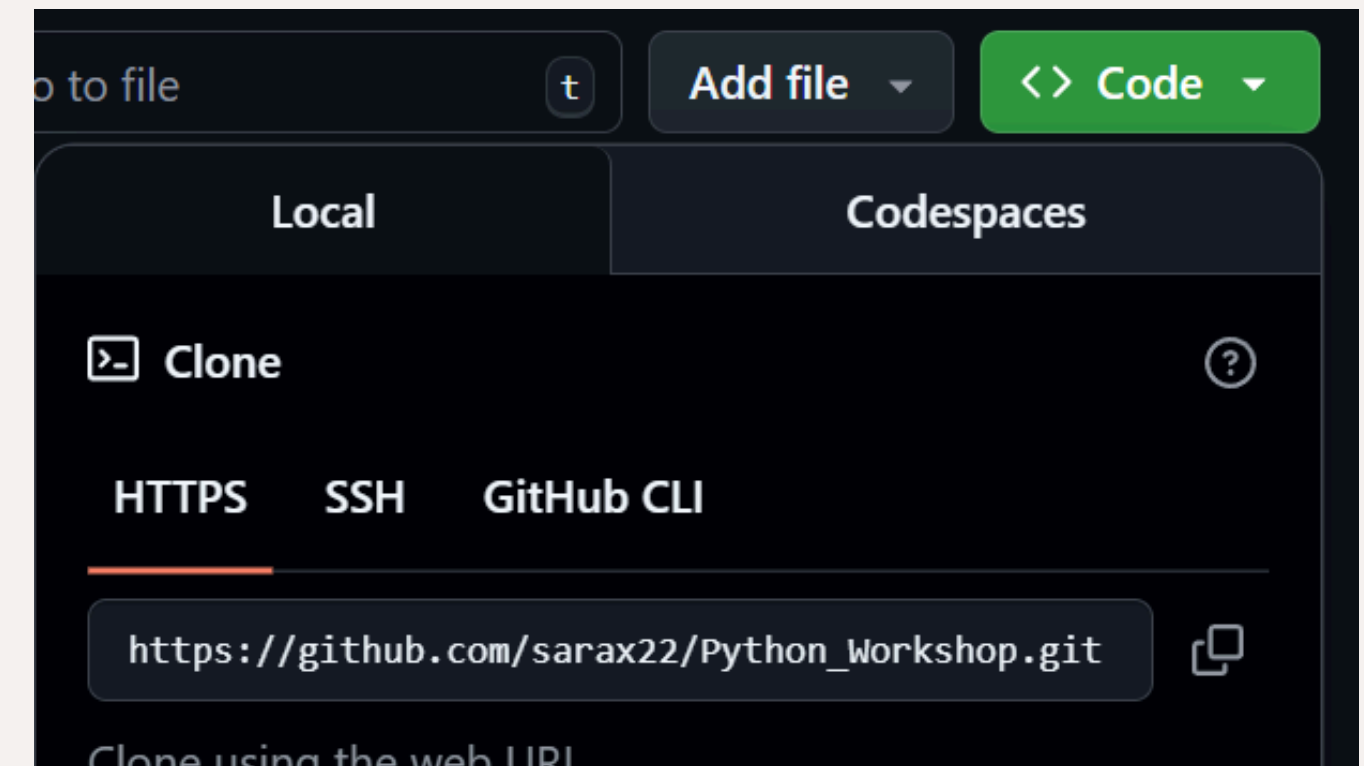


Cloning a Git Repository

- Go to https://github.com/sarax22/Python_Workshop
- Press the green <> Code Button
- Copy the HTTPS url
- Open VSCode
- Press “Clone Git Repository”
- Enter URL



What is MicroPython

- **MicroPython is a lightweight version of Python designed for microcontrollers**
- **It allows you to write Python code to control hardware devices like:**
 - **LEDs**
 - **sensors**
 - **motors**
- **Runs on boards such as:**
 - **Raspberry Pi Pico**
 - **ESP32**

MicroPython

Uses the **SAME** syntax as Python

E.g. Code Snippet that Turns an LED on:

```
from machine import Pin
led = Pin(25, Pin.OUT)
led.on()
```

E.g. Code snippet that blinks an LED every second

```
from machine import Pin
import time

led = Pin(25, Pin.OUT)

while True:
    led.toggle()
    time.sleep(1)
```

Tasks for Today:

- **Learn basic Python for hardware-style problems**
- **Think like an embedded system**
- **Work through mini tasks**
- **Build a robot pet (final challenge)**

BASICS

```
print("Hello, World!")      # Print output
x = 10                      # Variable assignment
type(x)                    # Get type of variable
input("Enter value: ")     # Take user input
```

DATA TYPES

```
my_list = [1, 2, 3]        # List
my_tuple = (1, 2, 3)       # Tuple
my_set = {1, 2, 3}         # Set
my_dict = {"key": "value"} # Dictionary
```

CONDITIONALS

```
if x > 5:
    print("Greater than 5") # If condition
elif x == 5:
    print("Equals 5")      # Else-if
else:
    print("Less than 5")   # Else
```

LOOPS

```
for i in range(5):
    print(i)                # For loop

while x > 0:
    print(x)
    x -= 1                 # While loop
```

Link to another [cheatsheet](#)